

B.Sc. (Hons) Semester II (NEP) Examination 2024–25
COMPUTER SCIENCE

Paper Code: CSCSE21 : Mobile Application Development

Time: 2:30 Hours

Full Marks: 70

Note: Question No. 1 is compulsory. Answer any FIVE from the remaining questions.

Question 1 (Compulsory)

Choose the correct option from the following:

[10 × 2 = 20]

- (a) What is Android?
 - (i) A mobile device (ii) An operating system (iii) A web browser (iv) A programming language
- (b) Which file is the entry point for an Android application configuration?
 - (i) `MainActivity.kt` (ii) `AndroidManifest.xml` (iii) `strings.xml` (iv) `build.gradle`
- (c) What does APK stand for?
 - (i) Android Phone Kit (ii) Android Package Kit (iii) Android Primary Key (iv) Application Package Key
- (d) Which language is officially recommended by Google for Android development since 2019?
 - (i) Kotlin (ii) C# (iii) JavaScript (iv) Ruby
- (e) What is the purpose of the `R.java` file?
 - (i) To store app settings (ii) To manage database (iii) To link XML resources with Java/Kotlin code (iv) To debug code
- (f) Which component is used to display a UI screen in Android?
 - (i) Broadcast Receiver (ii) Service (iii) Activity (iv) Content Provider
- (g) Which lifecycle method is called when an Activity is first created?
 - (i) `onStart()` (ii) `onCreate()` (iii) `onResume()` (iv) `onRestart()`
- (h) Which layout arranges elements linearly in a single column or row?
 - (i) `ConstraintLayout` (ii) `GridLayout` (iii) `FrameLayout` (iv) `LinearLayout`
- (i) Which of the following is used for local relational data storage in Android?
 - (i) Firebase (ii) MySQL (iii) SQLite (iv) PostgreSQL
- (j) What component allows sharing of data between different applications in Android?
 - (i) Content Provider (ii) Intent (iii) Service (iv) Fragment

Descriptive Questions (Attempt any FIVE)

1. Define any five of the following terms: [5 × 2 = 10]
 - (a) APK
 - (b) Gradle in Android Studio
 - (c) Layout XML
 - (d) `AndroidManifest.xml`
 - (e) Activity
 - (f) Intent
2. Write short answer type questions: [2 × 5 = 10]
 - (a) Describe the major architectural components of Android application development.
 - (b) Explain the use of SQLite in Android with an example.
3. Explain any five UI components (Buttons, TextView, EditText, ImageView, RecyclerView) in Android app development. [10]
4. Content Providers: [2 × 5 = 10]
 - (a) What is a Content Provider in Android? What operations does it perform?
 - (b) Explain the architecture and working mechanism of Content Providers.
5. Write SQLite queries for the following requirements: [5 × 2 = 10]
 - (a) Create a **student** table with attributes: **ID**, **name**, **age**, **gender**, **department**, and **marks**.
 - (b) Write a query to insert a record: Name: 'Raj', Age: 21, Gender: 'Male', Dept: 'Computer Science', Marks: 85.
 - (c) Write a query to display names of all students from the **student** table.
 - (d) Write a query to display all student records where marks > 80.
 - (e) Write a query to update marks of student with ID = 3 to 90.
6. Kotlin Programming: [2 × 5 = 10]
 - (a) Write a Kotlin function `add(a: Int, b: Int): Int` that returns the sum of two numbers.
 - (b) Write a Kotlin program to print numbers from 1 to 10 using a loop statement.